A Two-Thirds Hyperbolicity Wedge for Jensen Polynomials of Riemann's Xi-Function

John Savva

Independent researcher

Version 1.1 — 3 September 2026

Permanent record: `10.5281/zenodo.22293642`

Abstract

Let $\gamma(n)$ be defined by

$$\xi\left(\frac{1}{2}+w\right)=\sum_{n\geq 0}\frac{\gamma(n)}{n!}w^{2n},\qquad \mathcal{J}^{d,n}(X)=\sum_{j=0}^{d}\binom{d}{j}\gamma(n+j)X^{j}.$$

We prove that there is an absolute constant $K>0$ such that $n^2\log(n+2)\geq Kd^3$ implies that $\mathcal{J}^{d,n}$ has d distinct negative real zeros. The exponent $2/3$ comes from matching six consecutive coefficients with a four-parameter product of two Jacobi factors and then bounding the sixth interpolation remainder. The proof combines a direct sectorial saddle analysis of the xi coefficients, finite-free convolution, a non-perturbative hypergeometric recurrence, and complex Hermite–Genocchi interpolation. The argument does not prove the Riemann hypothesis and does not locate zeros of $\zeta(s)$.

AI-assistance disclosure

These papers and the associated Lean 4 formalization were developed with substantial artificial-intelligence assistance, primarily from **OpenAI GPT-5.6 Sol Pro**. Additional models assisted with adversarial testing, source comparison, independent recalculation, and AI-only technical review. **Wolfram Mathematica 15.0.1** was used in a clean-room computation executed by the author to check selected symbolic identities and exact algebraic calculations. Those computations provide independent-CAS corroboration of the specified calculations; they do not constitute a proof of the entire theorem, human mathematical review, or peer review. The author directed, reviewed, and edited the work, authorized the repository changes, and accepts responsibility for the contents.

Verification and review status

The accompanying formal record kernel-checks the complete T1–T5 analytic chain, the exact xi branch through six matches, and the concrete xi multiplier and final sign-transfer assembly. Its headline xi theorem is conditional only on explicitly typed Jacobi, MMP, and MSS literature inputs. Exact SymPy, a user-executed clean-room Mathematica reconstruction, and directed Arb/ACB calculations provide separate computational corroboration. The AI-assistance disclosure above applies to the paper and formal record. A local Palomar candidate covers the literal T5 statement, but official Comparator and NanoDa replay has not yet occurred.

2020 Mathematics Subject Classification. Primary 11M26; Secondary 26C10, 33C45, 68V15.

Keywords. Riemann xi-function, Jensen polynomials, hyperbolicity, Jacobi polynomials, finite free convolution, saddle-point asymptotics, Lean.

Contents

1 Introduction and theorem

The Jensen polynomials attached to a sequence $(a(n))_{n\geq 0}$ are

$$J_a^{d,n}(X) = \sum_{j=0}^{d} \binom{d}{j} a(n+j)X^j.$$

When $a(n) = \gamma(n)$ comes from Riemann's xi-function, hyperbolicity of these polynomials is a finite shadow of the real-zero geometry of the completed function. Griffin, Ono, Rolen, and Zagier proved eventual hyperbolicity for each fixed degree and established a Hermite limit [2]; Griffin, Ono, Rolen, Thorner, Tripp, and Wagner developed the xi-specific asymptotics [1]. Holland introduced a finite-free comparison architecture that yields a degree growing with the shift [3]. The present argument changes the local comparison from a five-match to a six-match model and replaces the portion of the old localization argument whose small-gap hypothesis fails on the new branch.

Theorem 1.1 (Two-thirds hyperbolicity wedge)**.** *There is an absolute constant $K > 0$ such that, for all integers $d \geq 1$ and $n \geq 0$,*

$$n^2 \log(n+2) \geq Kd^3 \tag{1.1}$$

implies that $\mathcal{J}^{d,n}$ has exactly d distinct negative real zeros.

The proof is developed in Sections 4–16 and is closed in an explicit *Proof of Theorem 1.1* in Section 16. The intermediate coefficient theorem, Theorem 7.1, is likewise followed immediately by its proof.

The constant is effective in the usual proof-theoretic sense: every finite geometric contribution is explicit, while two analytic constants can be read from the sectorial saddle proof. The direct unweighted sup-norm contraction is extremely wasteful, so we make no claim that the resulting constant is useful for computation. In particular, the numerical root examples in the verification package are checks, not the finite verification of an optimized threshold.

1.1 What the theorem does not say

Theorem 1.1 is a statement about finite polynomials formed from Taylor coefficients of ξ. It neither proves that the nontrivial zeros of ζ lie on the critical line nor sharpens a zero-free region. It also does not say that every Jensen polynomial is hyperbolic; it gives a sufficient asymptotic wedge. The theorem's scale is $d \ll (n^2 \log n)^{1/3}$, hence the name "two-thirds wedge" when the relation is expressed as a lower bound on n in terms of d.

1.2 Architecture

The proof has six interfaces.

1. The centered xi coefficients are expressed as a positive Mellin moment with the exact factor-eight normalization.

2. A sectorial saddle theorem gives an explicit nonzero main term and a holomorphic relative error, which can be differentiated six times.

3. Four quotient coordinates and two normalizations select a positive four-parameter hypergeometric comparison matching six coefficients.

4. Jacobi factorization and finite-free convolution give positive, simple, localized comparison roots.

5. A direct shifted ${}_3F_2$ recurrence bounds all derivatives at every critical point without treating the surviving gap parameter as small.

6. Complex six-node interpolation bounds the coefficient multiplier by $d^3/(n^2 \log n)$, after which sign transfer finishes the proof.

1.3 How to read the proof and the evidence labels

The manuscript numbering and the evidence-ledger numbering serve different purposes. Labels such as Theorem 7.1 are ordinary section-based theorem numbers. Labels **T1**–**T18** are stable identifiers for proof interfaces: some identify displayed theorems, while others identify a calculation or implication occupying a section. In particular, Theorem 7.1 is **T5**, and Theorem 1.1 is **T18**.

Evidence IDs	Proof block and primary manuscript location
T1–T6	Xi coefficients and saddle: Sections 4–8; Appendices B–E
T7–T9	Comparison branch: Sections 9–11; Appendix F
T10–T12	Finite-free root geometry: Section 12; Appendix G
T13–T14	Recurrence and radius: Sections 13–14; Appendix H
T15–T18	Residual and conclusion: Sections 15–17; Appendices I–J

The theorem-to-evidence appendix and the separate machine-readable assurance matrix then say which parts are paper analysis, which implications are kernel checked, and which computations merely corroborate the proof. They are navigation and trust-boundary documents, not substitute proofs.

The logical dependencies are directional. Numerical evidence never enters a proof premise, and Holland's headline theorem is not used. The imported finite-free results are the positive-root and log-mesh statements of Martínez-Finkelshtein–Morales–Perales [5], together with the maximum-root product inequality of Marcus–Spielman–Srivastava [4, Theorem 1.6].

2 Definitions and normalization

We use

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

It is entire and satisfies $\xi(1-s) = \xi(s)$. We work with the complex centered variable w, not the classical real variable t in $\Xi(t) = \xi(1/2 + it)$. Thus no hidden power of i enters the coefficient sequence.

Definition 2.1 (Xi coefficients and Jensen polynomial)**.** The real coefficients $\gamma(n)$ and the Jensen polynomial are defined by

$$\xi(1/2 + w) = \sum_{n\geq 0} \frac{\gamma(n)}{n!} w^{2n}, \tag{2.1}$$

$$\mathcal{J}^{d,n}(X) = \sum_{j=0}^{d} \binom{d}{j} \gamma(n+j) X^j. \tag{2.2}$$

The functional equation makes the left side of (2.1) even. Equivalently,

$$\gamma(n) = \frac{n!}{(2n)!} \frac{\mathrm{d}^{2n}}{\mathrm{d}w^{2n}} \xi(1/2 + w)\bigg|_{w=0}. \tag{2.3}$$

This normalization is load-bearing. The two most dangerous transcription errors are replacing $n!$ by 2^{2n} and replacing the kernel weight $e^{u/2}$ below by e^u; both give plausible-looking but false formulas.

2.1 Comparison variables

Write

$$x = n^{-1}, \qquad e = \mathcal{L}_n^{-1}, \qquad \mathcal{L}_n = L_{2n-2},$$

where L_N is the saddle variable defined in Section 5. For positive parameters $y = (\alpha, t, w, \delta)$ set

$$A = \frac{\alpha}{xe}, \qquad B = \frac{t + we}{x}, \qquad C = \frac{t}{x}, \qquad D = \frac{1 + \delta e}{x}. \tag{2.4}$$

The comparison polynomial is

$$p_F(y_0) = {}_3F_2\left(\begin{matrix} -d, A, C \\ B, D \end{matrix}; \frac{D}{AC} y_0\right). \tag{2.5}$$

The independent variable is denoted y_0 in this display to distinguish it from the parameter vector y.

3 Prior work and precise novelty

GORZ expand logarithmic coefficient ratios and obtain a Hermite limit for each fixed degree [2]. GORTTW supply the xi moment asymptotics needed when the degree varies [1]. Holland replaces a local Hermite perturbation by a hypergeometric comparison formed with finite-free convolution [3]. We retain three parts of that architecture: a low-dimensional matching system, positive-rooted Jacobi factors, and sign stability at critical points.

The change from the earlier three-fifths scale to (1.1) is local and exact. Matching six coefficients makes the logarithmic ratio vanish at six consecutive nodes. The six-node Hermite–Genocchi remainder contains six factors, and the critical-point radius is $\rho \asymp \sqrt{nd}$. Therefore

$$\frac{\rho^6}{n^5 \log n} \asymp \frac{(nd)^3}{n^5 \log n} = \frac{d^3}{n^2 \log n}.$$

The wedge is precisely the condition that this quantity be small.

Three repairs are essential rather than cosmetic.

1. The exact xi/Mellin identity is derived directly, because two printed coefficient conventions differ by a factor of eight.
2. Holland's assembled localization lemma is not imported: its $(C - D)/D \le 1/4$ hypothesis fails on the new branch, where the quotient tends to 1.

3. The shifted hypergeometric recurrence retains $\epsilon_p = (C - D)/C$, which tends to 1/2 and is not perturbative.

The resulting proof shares an architecture with the cited work but not a black-box dependence on its headline conclusion.

4 Exact xi/Mellin identity

Let

$$\Theta(t) = \sum_{m\geq 1} e^{-\pi m^2 t}, \qquad \omega(t) = \frac{1}{2}\big(2t^2\Theta''(t) + 3t\Theta'(t)\big).$$

Riemann's integral representation, integration by parts, and the change of variables $t = e^{2u}$ give

$$\xi(1/2 + w) = 8\int_0^\infty \omega(e^{2u})e^{u/2}\cosh(wu)\,\mathrm{d}u. \tag{4.1}$$

The kernel is positive for $u > 0$, decays superexponentially as $u \to +\infty$, and extends to an even full-line kernel by the modular transformation of the theta function. These properties justify termwise differentiation on compact w-sets.

Proposition 4.1 (Factor-eight coefficient identity)**.** *For every integer $n \geq 0$,*

$$\gamma(n) = \frac{8n!}{(2n)!}\int_0^\infty \omega(e^{2u})e^{u/2}u^{2n}\,\mathrm{d}u. \tag{4.2}$$

Proof. Expand $\cosh(wu) = \sum_{n\geq 0}(wu)^{2n}/(2n)!$ in (4.1). Dominated convergence on compact w-sets permits coefficient extraction. Comparing with (2.1) gives (4.2). The duplication formula yields the equivalent prefactor

$$\frac{8n!}{(2n)!} = \frac{8\sqrt{\pi}}{4^n\Gamma(n + 1/2)}.$$

□

Expanding Θ mode by mode introduces an auxiliary Mellin moment $F(s)$. With $N = 2M - 2$, direct algebra gives the exact continued identity

$$\gamma(M) = \frac{\Gamma(M+1)}{\Gamma(2M+1)}2^{-2M-2}\left(32\binom{2M}{2}F(2M-2) - F(2M)\right). \tag{4.3}$$

It is initially obtained in a right half-plane where the integrals converge absolutely and then continued holomorphically. The relation $32\binom{N+2}{2} = 16(N+2)(N+1)$ will be used in the saddle assembly. For later use define the continued moment

$$M_z := 2^{-2z-2}\left(32\binom{2z}{2}F(2z-2) - F(2z)\right). \tag{4.4}$$

Then (4.3) is exactly

$$\gamma(z) = \frac{\Gamma(z+1)}{\Gamma(2z+1)}M_z. \tag{4.5}$$

5 Quantitative sectorial saddle

Fix principal logarithms in a sector $|\arg N| \le \theta < \pi/2$ and define

$$N = L\left(\pi e^{L} + \frac{3}{4}\right) =: \Psi(L). \tag{5.1}$$

Put $\ell = \log|N|$ and

$$L_0 = \log N - \log\log N - \log \pi.$$

Lemma 5.1 (Sectorial saddle branch). *For $|N| \ge e^{12}$ in a fixed sector of angle less than $\pi/2$, there is a unique zero L_N of* (5.1) *in $|L - L_0| < 1$. These zeros patch to a holomorphic branch agreeing with the positive real solution. On every fixed smaller closed sector,*

$$|L_N| \asymp \log|N|, \qquad r = L_N^{-1} \to 0, \qquad \sigma = L_N/N \to 0, \tag{5.2}$$

$$|L_N| \ge \tfrac{1}{2}\log|N|, \qquad |r|, |\sigma| \le \frac{7}{50}, \tag{5.3}$$

$$Q_N = (1 + L_N)N - \tfrac{3}{4}L_N^2 = N L_N(1 + r - \tfrac{3}{4}\sigma), \tag{5.4}$$

$$|1 + r - \tfrac{3}{4}\sigma| \ge \frac{9}{16}, \qquad L_N' = \frac{L_N}{Q_N}. \tag{5.5}$$

Proof. On $D_N = \{L : |L - L_0| \le 1\}$ consider

$$G_N(L) = L + \log L + \log \pi - \log\left(1 - \frac{3L}{4N}\right) - \log N.$$

Let

$$m(\ell) = \ell - \log(\ell + 1) - \log \pi - 1, \quad M_\theta(\ell) = \ell + 2 + \log(\ell + 1) + \theta/\ell + \log \pi.$$

For $L \in D_N$, direct geometry gives $\Re L \ge m(\ell) > 0$ and $|L| \le M_\theta(\ell)$. Hence every logarithm is legal, and $|3L/(4N)| < 10^{-4}$ when $\ell \ge 12$.

On ∂D_N, compare G_N with $H_N(L) = L - L_0$. The inequality $|\log(1 + z)| \le 2|z|$ for $|z| \le 1/2$ gives

$$|G_N - H_N| \le 2\frac{\log(\ell + 1) + \theta/\ell + \log \pi + 1}{\ell} + \frac{3M_\theta(\ell)}{2e^{\ell}} < 1. \tag{5.6}$$

Rouché's theorem yields one zero, counted with multiplicity. Exponentiating $G_N(L_N) = 0$ gives (5.1). The derivative of the left side of (5.1) is Q_N/L_N. Before invoking the implicit-function theorem, use (5.3) and the reverse triangle inequality:

$$|1 + r - \tfrac{3}{4}\sigma| \ge 1 - \frac{7}{50} - \frac{3}{4}\frac{7}{50} = \frac{151}{200} > \frac{9}{16}.$$

Thus $Q_N \ne 0$, the zero is simple, and local holomorphic branches agree on overlaps by uniqueness. Conjugation symmetry and uniqueness on the positive axis select the required branch. The disc bounds also give $m(\ell) > \ell/2$ and the limits in (5.2). Implicit differentiation now gives (5.5). ∎

The reduced denominator needed at order six has the stronger explicit bound

$$|4 + 4r - 3\sigma| \geq 4 - 4\frac{7}{50} - 3\frac{7}{50} = \frac{151}{50}. \tag{5.7}$$

6 Contour deformation and leading mode

For the leading theta mode, write the Mellin integral in $u = \log t$ and keep the Jacobian outside the phase so that the saddle remains (5.1). Explicitly, set

$$\Phi_s(u) = s \operatorname{Log} u - \frac{3}{4}u - \pi e^u, \qquad g_s(u) = e^u e^{\Phi_s(u)}, \qquad I_1(s) := F_1(s) := \int_0^\infty g_s(u)\, \mathrm{d}u.$$

Write $L_s = \alpha + ib$. Split the original ray at $u = 1$ and apply Cauchy's theorem to the rectangle with vertices $1, X, X + ib, 1 + ib$. The right connector tends to zero as $X \to \infty$, giving the exact identity

$$I_1(s) = E_s + \int_1^\infty g_s(x + ib)\, \mathrm{d}x, \qquad E_s = \int_0^1 g_s(x)\, \mathrm{d}x + i\int_0^b g_s(1 + iy)\, \mathrm{d}y.$$

Thus only the ray $[1, \infty)$ is translated. It remains in $\Re u \geq 1$, away from the principal-log cut. On this actual ray put $u = L_s + r$, so $r \geq 1 - \Re L_s$; the expression $L_s + iv$ would be vertical and would reverse the quadratic sign. The exact phase difference has quadratic part $-K_s r^2/2$, where

$$K_s = \frac{s}{L_s}\left(1 + \frac{1}{L_s} - \frac{3L_s}{4s}\right), \qquad Q_s = L_s^2 K_s. \tag{6.1}$$

The direct imaginary-part bootstrap from the saddle equation gives $|\arg K_s| < 1/20$, and therefore

$$\Re K_s \geq \cos(1/20)|K_s| > 0. \tag{6.2}$$

Choose a central window $|r| \leq |K_s|^{-2/5}$. It lies on the actual ray for all sufficiently large $|s|$. Taylor's theorem with an integral remainder gives

$$\Phi_s(L_s + r) - \Phi_s(L_s) = -\frac{1}{2}K_s r^2 + \frac{1}{6}\Phi_s^{(3)}(L_s) r^3 + R_4(s, r),$$

with $|R_4(s, r)| \leq C|K_s||r|^4$ on the fixed sector. The Jacobian outside the phase contributes the exact amplitude e^r. After the omitted left Gaussian tail has been bounded exponentially, the full real Gaussian is used only as a comparison integral. Completing the square gives

$$\frac{\int_{\mathbb{R}} r^3 e^{r - K_s r^2/2}\, \mathrm{d}r}{\int_{\mathbb{R}} e^{r - K_s r^2/2}\, \mathrm{d}r} = \frac{3}{K_s^2} + \frac{1}{K_s^3}.$$

Since $\Phi_s^{(3)}(L_s) = O(|K_s|)$, the signed cubic therefore contributes $O(|K_s|^{-1})$ relatively; estimating its absolute value first would lose this order. Equation (6.2) dominates the central integrand by a real Gaussian, so dominated convergence is uniform.

Outside the central window, strict concavity on the actual shifted ray gives an increasing left tail on $1 \le x \le \alpha - V$ and an exponentially decreasing right tail on $x \ge \alpha + V$, where $V = |K_s|^{-2/5}$. The two tail integrals are $O(e^{-c|K_s|^{1/5}})$ relative to the saddle main term. The endpoint term E_s satisfies the stronger $O(e^{-c|s|\log\log|s|})$ relative bound. Consequently the leading moment has

$$F_1(s) = \mathcal{A}(s)\left(1 + O\left(\frac{1}{|K_s|}\right)\right), \tag{6.3}$$

where

$$\mathcal{A}(s) = \sqrt{\frac{2\pi}{K_s}}L_s^s \exp\left(\frac{L_s}{4} - \frac{s}{L_s} + \frac{3}{4}\right). \tag{6.4}$$

All roots and logarithms in (6.4) are continued from the positive axis. The detailed connector parametrizations and numerical-free constants are recorded in the technical supplement.

7 Higher theta modes and coefficient theorem

For $N = 2M - 2$, define the candidate coefficient main term by

$$\mathcal{G}(M) = \frac{e^{M-2}M^{M+1/2}L_N^N}{2^{2M-2}N^{N+1/2}}\sqrt{\frac{2\pi}{K_N}}\exp\left(\frac{L_N}{4} - \frac{N}{L_N} + \frac{3}{4}\right), \tag{7.1}$$

where all powers and the square root are continued from the positive axis.

Theorem 7.1 (Sectorial coefficient asymptotic). *Let $\theta_0 = 1/400$ and $\theta_1 = 1/200$. There are $R, C > 0$ such that $\mathcal{G}$ is holomorphic and nonzero on*

$$|M| > R, \qquad |\arg M| < \theta_1,$$

and

$$\gamma(M) = \mathcal{G}(M)(1 + \mathcal{E}(M)), \qquad |\mathcal{E}(M)| \le C\frac{\log|M|}{|M|} \tag{7.2}$$

on the closed inner sector $|\arg M| \le \theta_0$. The error $\mathcal{E}$ is holomorphic on the outer sector.

Proof. The mth theta mode has the same analytic contour and an extra exponential term proportional to $-(m^2-1)e^{L_s+r}$. On the central window its real part is at most $-c(m^2 - 1)|s|/|L_s|$; on the tails the leading concavity estimate already dominates. Summation over $m \ge 2$ is therefore exponentially small compared with (6.3). This conclusion is uniform on the fixed inner sector, not uniform in an arbitrary free sector angle.

Insert (6.3) into (4.3). A two-step Taylor expansion of $\log\mathcal{A}(s)$ controls $F(N+2)/F(N)$, and sectorial Stirling controls the gamma ratio. The cancellation

$$\frac{\mathrm{d}}{\mathrm{d}s}\log\mathcal{A}(s) = \log L_s + \frac{1}{L_sK_s} - \frac{1}{2}\frac{K_s'}{K_s} \tag{7.3}$$

uses the saddle equation exactly. With $N = 2M - 2$ the main term is (7.1). More explicitly, Taylor's formula on the segment from N to $N + 2$ and the bound for the second derivative of $\log \mathcal{A}$ give

$$\frac{F(N+2)}{F(N)} = L_N^2 \left(1 + O\left(\frac{\log|N|}{|N|}\right)\right).$$

Combining this ratio with (4.3), the leading-mode estimate (6.3), the exponentially small sum of the higher modes, and sectorial Stirling yields (7.2).

The shrink from the leading-mode sector ensures that $N = 2M-2$ remains in the required domain after increasing R. The fixed angles also leave room for proportional Cauchy discs around points in the inner sector. The saddle lemma and (6.2) make every factor of (7.1) holomorphic and nonzero there after continuation from the positive axis. Hence the quotient defining $\mathcal{E}$ is holomorphic on the outer sector; the preceding uniform estimates give the asserted bound on the closed inner sector. Appendix D records the connector, central-window, tail, mode-summation, and two-step-ratio estimates separately.

The accompanying Lean theorem follows these quantifiers literally. It uses the wider proof sector $|\arg s| < 1/100$ to transport $N = 2M - 2$, proves holomorphy and nonvanishing on $|\arg M| < 1/200$, and proves the displayed rate on $|\arg M| \leq 1/400$, with explicit witnesses for R and C. Its Cauchy corollary transports the resulting error through order six. □

8 Logarithmic derivatives through order six

By (4.5), Theorem 7.1, and the nonvanishing of the gamma function, M_z is nonzero after increasing the sectorial threshold. Let

$$h(z) := \log M_z \tag{8.1}$$

on the branch real on the positive axis. This is the logarithm of the auxiliary moment, not $\log \gamma(z)$. The exact bridge is

$$\log \gamma(z) = h(z) + \log \Gamma(z+1) - \log \Gamma(2z+1),$$

with branches continued from positivity. Theorem 7.1 supplies a holomorphic relative error on a larger sector. If a disc of radius R_x around x remains in that sector and $|\mathcal{E}| \leq \epsilon_x$ on its boundary, Cauchy's estimate gives

$$|\mathcal{E}^{(j)}(x)| \leq \frac{j!\epsilon_x}{R_x^j}. \tag{8.2}$$

Taking R_x proportional to x transfers the error in (7.2) through every derivative needed below.

Before differentiating, convert the coefficient main term to the auxiliary-moment main term by subtracting the explicit gamma factors in the exact bridge:

$$\log \mathcal{G}(x) - \log \Gamma(x+1) + \log \Gamma(2x+1).$$

After the sectorial Stirling cancellations already used to obtain (7.1), its moving-saddle logarithm is

$$G_0(N) = (N+1)\log L_N + \frac{L_N}{4} - \frac{N}{L_N} - \frac{1}{2}\log Q_N, \tag{8.3}$$

with logarithms continued from the positive axis. Put $N = 2x-2$, $r = L_N^{-1}$, and $\sigma = L_N/N$. Total implicit differentiation of this moment saddle term, including the chain $N = 2x-2$, yields

$$\begin{aligned}
h''(x) &= \frac{2}{xL_N}\big(1+O(L_N^{-1})\big),\\
h^{(3)}(x) &= -\frac{2}{x^2L_N}\big(1+O(L_N^{-1})\big),\\
h^{(4)}(x) &= \frac{4}{x^3L_N}\big(1+O(L_N^{-1})\big),\\
h^{(5)}(x) &= -\frac{12}{x^4L_N}\big(1+O(L_N^{-1})\big),\\
h^{(6)}(x) &= \frac{48}{x^5L_N}\big(1+O(L_N^{-1})\big).
\end{aligned} \tag{8.4}$$

Before the chain rule, the fifth and sixth leading derivatives have constants -6 and 24; after $N = 2x-2$ they become -12 and 48. Keeping both stages visible prevents a factor-two ambiguity.

Its exact sixth derivative is

$$G_0^{(6)}(N) = \frac{H_6(r,\sigma)}{N^5L_N}, \qquad H_6(r,\sigma) = \frac{P_{13}(r,\sigma)}{(4+4r-3\sigma)^{12}}, \tag{8.5}$$

where P_{13} has total degree 13 and 82 nonzero monomials. Exact coefficientwise majorization on $|r|, |\sigma| \le 7/50$ gives

$$|H_6(r,\sigma)| \le \frac{6422139805764931584036533551104}{70257609972813759418868 4005} < 10^4. \tag{8.6}$$

Together with (5.7), this yields the deliberately coarse bound

$$|G_0^{(6)}(N)| \le \frac{20000}{|N|^5\log|N|}. \tag{8.7}$$

The exact rational tower is reconstructed independently by SymPy, Mathematica, and ACB power series. The paper does not ask a reader to trust a decimal fit for any constant in (8.4).

9 Six matches and quotient coordinates

The comparison coefficients are chosen through four consecutive quotient coordinates. If (a_j) is a nonzero sequence with compatible logarithms, define

$$q_k(a) = \log a_{k+2} - 2\log a_{k+1} + \log a_k, \qquad 0 \le k \le 3. \tag{9.1}$$

These are logarithms of the ratios $a_{k+2}a_k/a_{k+1}^2$.

Lemma 9.1 (Four quotients plus two normalizations)**.** *Suppose $a_j, b_j \neq 0$ for $0 \leq j \leq 5$, logarithms have been chosen compatibly, $q_k(a) = q_k(b)$ for $0 \leq k \leq 3$, and $a_0 = b_0$, $a_1 = b_1$. Then $a_j = b_j$ for $0 \leq j \leq 5$.*

Proof. Set $c_j = \log a_j - \log b_j$. The hypotheses say $c_0 = c_1 = 0$ and $c_{k+2} = 2c_{k+1} - c_k$ for $0 \leq k \leq 3$. Induction gives $c_2 = \cdots = c_5 = 0$. Exponentiation gives the result. □

For the xi coefficients and the hypergeometric model, the four equations are written as a residual map $G_n(\alpha, t, w, \delta) = 0$. The logarithmic gamma terms are decomposed before estimating. For

$$f_k(U) = \log(U + k) - \log(U + k + 1),$$

the exact half-shift is $-f_k(n + 1/2)$. Pairing it with the nearby D term prevents an unnecessary factor of L_n.

10 Elementary cube calculus

For $q = j + 1$, repeated real FTC gives

$$\Delta^j f_0(U) = (-1)^{j+1} j! \int_{[0,1]^q} (U + u_1 + \cdots + u_q)^{-q} \, \mathrm{d}u. \tag{10.1}$$

Define

$$\Phi_q(s, z) = \int_{[0,1]^q} (s + z(u_1 + \cdots + u_q))^{-q} \, \mathrm{d}u. \tag{10.2}$$

The finite-dimensional cube has volume one. If $s \geq s_0 > 0$ and $z \geq 0$, differentiation under the integral gives, for $r = 0, 1, 2$,

$$\partial_s^r \Phi_q(s, z) = (-1)^r (q)_r \int_{[0,1]^q} (s + z \textstyle\sum u_i)^{-q-r} \, \mathrm{d}u, \tag{10.3}$$

$$|\partial_s^r \Phi_q(s, z)| \leq (q)_r s_0^{-q-r}. \tag{10.4}$$

The mean-value theorem yields the explicit first-order comparison

$$\left|\partial_s^r \Phi_q(s, z) - (-1)^r (q)_r s^{-q-r}\right| \leq (q)_r (q + r) q z \, s_0^{-q-r-1}. \tag{10.5}$$

The normalized elementary part of the four-coordinate residual is

$$\begin{aligned} H_{n,j} = \frac{a_j}{e} \{ & e^q \Phi_q(\alpha, xe) - \Phi_q(t + we, x) + \Phi_q(t, x) \\ & - \Phi_q(1 + \delta e, x) + \Phi_q(1 + x/2, x)\}, \end{aligned} \tag{10.6}$$

with

$$(a_0, a_1, a_2, a_3) = (1, 1/2, 1, 1). \tag{10.7}$$

We record the three different limiting mechanisms, since conflating them is a common source of a false scale.

1. The remote A boundary contains the external factor e^q/e and therefore contributes $1/\alpha$ only for $q = 1$.
2. The $B - C$ divided difference has length we and tends to qw/t^{q+1}.
3. The D–gamma half-shift pair has length $\delta e - x/2$ after the correct rescaling and tends to $q\delta$.

Applying (10.4)–(10.5) to both values and first parameter derivatives gives uniform convergence on the rational parameter box:

$$H^\infty(\alpha, t, w, \delta) = \left(\frac{1}{\alpha} + \frac{w}{t^2} + \delta, \frac{w}{t^3} + \delta, \frac{3w}{t^4} + 3\delta, \frac{4w}{t^5} + 4\delta\right). \tag{10.8}$$

The value error is $O(e + x/e)$ and the derivative error is $O(e + x)$. The xi contribution from (8.4) is

$$S^\infty = (-2, -1, -2, -2). \tag{10.9}$$

The estimates in this section are finite-dimensional and elementary. Their two Fubini orientations, derivative identities, parameter segments, and remote-boundary constants are all kernel-checked in the accompanying Lean modules.

11 Quantitative positive branch

The limiting map $F = H^\infty + S^\infty$ has the zero

$$y_* = (\alpha, t, w, \delta) = (3, 2, 16/3, 1/3). \tag{11.1}$$

Writing F_j for the four components of $F = H^\infty + S^\infty$, exact elimination gives

$$3F_1 - F_2 = \frac{3w(t-1)}{t^4} - 1, \qquad \frac{4}{3}F_2 - F_3 = \frac{4w(t-1)}{t^5} - \frac{2}{3}. \tag{11.2}$$

At a zero, these identities give $3w(t-1) = t^4$ and $6w(t-1) = t^5$; positivity forces $t = 2$, and then successively $w = 16/3$, $\delta = 1/3$, and $\alpha = 3$. Thus the positive-orthant solution is unique. In the coordinate order (α, t, w, δ),

$$DF(y_*) = \begin{pmatrix} -1/9 & -4/3 & 1/4 & 1 \\ 0 & -1 & 1/8 & 1 \\ 0 & -2 & 3/16 & 3 \\ 0 & -5/3 & 1/8 & 4 \end{pmatrix}, \tag{11.3}$$

and exact row reduction gives

$$\det DF(y_*) = -\frac{1}{144}, \qquad \left\|DF(y_*)^{-1}\right\|_\infty = \frac{304}{3}. \tag{11.4}$$

The outer parameter box is

$$\alpha \in [5/2, 7/2], \quad t \in [7/4, 9/4], \quad w \in [5, 6], \quad \delta \in [1/4, 5/12]. \tag{11.5}$$

Use the fixed inverse $P = DF(y_*)^{-1}$ and define $T_n(y) = y - PG_n(y)$. The uniform C^1 estimates above, after increasing the analytic threshold, give on a smaller sup-norm box

$$\|T_n(y_*) - y_*\|_\infty \le R/2, \qquad \sup_{\|y-y_*\|_\infty \le R} \|DT_n(y)\|_\infty \le 1/2. \tag{11.6}$$

The exact branch ledger verifies the rational box margins, the inverse row-sum arithmetic, and the implication from these two analytic inequalities; it does not purport to compute xi-specific residual or Jacobian enclosures. Banach's theorem gives a unique zero y_n in that box and the quantitative convergence

$$y_n - y_* = O(L_n^{-1}). \tag{11.7}$$

The word "unique" in this statement always means unique in the certified box; global uniqueness of G_n is neither needed nor claimed.

For $0 < e \le 1/12$, endpoint arithmetic on (11.5) gives

$$\frac{\alpha}{e} \ge 30 > \frac{11}{4} \ge t + we, \qquad t - 1 - \delta e \ge \frac{103}{144} > 0. \tag{11.8}$$

Hence (2.4) satisfies

$$A > B > C > D > 0. \tag{11.9}$$

12 Jacobi factors and finite-free convolution

For $s > 0$ put

$$Q^{(s)}_{U,V}(X) = {}_2F_1\left(\begin{matrix} -d, U \\ V \end{matrix}; \frac{sX}{U}\right).$$

Term-by-term comparison of the terminating series gives the exact identity

$$p_F = Q^{(1)}_{A,B} \boxtimes_d Q^{(D)}_{C,D} \tag{12.1}$$

in the ascending constant-term-one convention. Appendix G.2 displays the coefficient calculation and checks every hypothesis of the cited MMP propositions for these particular parameters. The literature uses the monic descending elementary-symmetric convention. If $p^\vee(X) = X^d p(1/X)/p(0)$, direct coefficient algebra gives

$$(p \boxtimes_d q)^\vee = p^\vee \boxtimes_d q^\vee. \tag{12.2}$$

Reversal reciprocates positive roots and preserves strict logarithmic mesh.

Order roots as $\lambda_1 \geq \cdots \geq \lambda_d > 0$ and define

$$\operatorname{lmesh}(p) = \min_{1 \leq j < d} \frac{\lambda_j}{\lambda_{j+1}} \geq 1. \tag{12.3}$$

The inequalities $A \geq B + d$, $C \geq D + d$, and $B, D > 0$ identify both factors in (12.1) with positive rescalings of Jacobi polynomials having parameters greater than -1. Thus each has degree d and d simple positive roots. After the explicit reversal adapter, MMP Proposition 2.7(iii) preserves nonnegative real-rootedness under multiplicative convolution, while Proposition 2.17 in the pinned v3 convention gives

$$\operatorname{lmesh}(p \boxtimes_d q) \geq \operatorname{lmesh}(p). \tag{12.4}$$

For $d \geq 2$, the first factor has logarithmic mesh strictly greater than one, so (12.4) makes the convolution simple. Proposition 2.7(iii) puts all of its roots in $[0, \infty)$, and its constant term is one, so zero is not a root. The cases $d = 0, 1$ are immediate from the constant and linear coefficients. Finally (12.1), rather than an assumption about p_F, transports this conclusion to the concrete ${}_3F_2$ comparison.

The Jacobi roots are localized without using Holland's inapplicable gap hypothesis. Transport the standard Jacobi matrix from $[-1, 1]$ to the positive-root variable. Direct diagonal and off-diagonal estimates give

$$U \geq V + d, \quad V \geq 32d \quad \Longrightarrow \quad \operatorname{roots}(q_{U,V}) \subset [V - 8\sqrt{Vd}, V + 8\sqrt{Vd}]. \tag{12.5}$$

The constant 8 is a Gershgorin consequence of a diagonal displacement at most $4\sqrt{Vd}$ and adjacent off-diagonal row sum at most $4\sqrt{Vd}$.

Before using the sharper product constant, the wedge and parameter box give the non-circular estimates

$$n \leq B \leq 3n, \quad n/2 \leq D \leq 2n, \quad A \geq 8B, \quad C \geq D + d, \quad B, D \geq 256d. \tag{12.6}$$

Theorem 1.6 of MSS bounds the maximum root of a multiplicative convolution by the product of maximum roots. Apply it both to the original polynomials and to their reversals. The reciprocal application requires positive lower endpoints. The preliminary geometry supplies them without circularity: from $B, D \geq 256d$,

$$\sqrt{Bd} \leq B/16, \qquad \sqrt{d/D} \leq 1/16,$$

and hence

$$B - 8\sqrt{Bd} \geq B/2 > 0, \qquad 1 - 8\sqrt{d/D} \geq 1/2 > 0.$$

These two strict inequalities are explicit arguments of the typed MSS interface and are proved in Lean before that interface is invoked. If u and v are the factor deviations from B and 1, then

$$|(B + u)(1 + v) - B| \leq 8\sqrt{Bd} + 8B\sqrt{d/D} + 64\sqrt{Bd}\sqrt{d/D}$$

$$\leq (12 + 8\sqrt{6})\sqrt{Bd}. \tag{12.7}$$

Thus every root, and by interlacing every critical point, lies in

$$|y_0 - B| \leq C_{\mathrm{loc}}\sqrt{Bd}, \qquad C_{\mathrm{loc}} = 12 + 8\sqrt{6} < 32. \tag{12.8}$$

The safe preliminary constant is $K_{\mathrm{pre}} = 256$, not 32; the latter would allow a negative lower endpoint in the ratio-free second factor.

13 Terminating hypergeometric ODE

For $0 \leq m \leq d$, let g_m be the shifted terminating polynomial with parameters

$$(m - d, A + m, C + m; B + m, D + m).$$

Its coefficient ratio is

$$\frac{c_{k+1}}{c_k} = \frac{D(k + m - d)(k + A + m)(k + C + m)}{AC(k + 1)(k + B + m)(k + D + m)}. \tag{13.1}$$

Cross multiplication is valid for the terminating finite producer under the positive parameter hypotheses. With $\vartheta = y_0 \,\mathrm{d}/\,\mathrm{d}y_0$, the coefficient identity gives the shifted Euler equation

$$\vartheta(\vartheta + B + m - 1)(\vartheta + D + m - 1)g_m = \frac{D}{AC} y_0(\vartheta + m - d)(\vartheta + A + m)(\vartheta + C + m)g_m. \tag{13.2}$$

The derivative-shift identity identifies g_m with $p_F^{(m)}$ up to a nonzero constant. Expanding (13.2) therefore gives

$$P_{3,m}T_{m+3} + P_{2,m}T_{m+2} + P_{1,m}T_{m+1} + P_{0,m}T_m = 0, \tag{13.3}$$

where

$$T_k = \frac{y_0^k p_F^{(k)}(y_0)}{p_F(y_0)}. \tag{13.4}$$

For compact formulas, set

$$\begin{aligned}
a &= 1 - y_0/A, \\
b_m &= B + m - y_0 + (d - 1 - 2m)y_0/A, \\
c_m &= (d - m)(1 + m/A)y_0, \\
\epsilon_p &= (C - D)/C, \\
\beta_m &= A(2m + 1 - d) - d(2m + 1) \\
&\quad + 3m^2 + 3m + 1, \\
\Gamma_m &= m(m - d)(m + A).
\end{aligned}$$

Then

$$\begin{aligned}
P_{3,m} &= a + \epsilon_p y_0/A = (AC - Dy_0)/(AC),\\
P_{2,m} &= b_{m+1} + (D+m)a + \epsilon_p(y_0/A)(A - d + 3 + 3m),\\
P_{1,m} &= c_{m+1} + (D+m)b_m + \epsilon_p(y_0/A)\beta_m,\\
P_{0,m} &= (D+m)c_m + \epsilon_p(y_0/A)\Gamma_m = \frac{Dy_0(A+m)(C+m)(d-m)}{AC}.
\end{aligned} \tag{13.5}$$

The finite coefficient producer, derivative shift, Euler ODE, and all four closed forms are proved in Lean. Mathematica independently returned four zero coefficient differences and verified every $0 \le m \le d$ for test degrees $5, 8, 11$.

14 Critical-point derivative radius

At a critical point of p_F, $T_0 = 1$ and $T_1 = 0$. On the localized interval (12.8), exact parameter inequalities give, uniformly for $0 \le m \le d-2$,

$$P_{2,m} \ge n/8, \quad |P_{3,m}| \le 2, \quad |P_{1,m}| \le C_1(n\sqrt{Bd} + nd), \quad 0 \le P_{0,m} \le C_0 n^2 d. \tag{14.1}$$

The constants may be taken as $C_0 = 48$ and $C_1 < 96$ after the safe geometric thresholds are imposed. Note that $\epsilon_p \to 1/2$; every occurrence in (13.5) remains in the main recurrence.

Choose $R = K_r\sqrt{Bd}$ with $K_r = 4096$ and put

$$\mathcal{M} = \max_{0 \le j \le d} \frac{|T_j|}{R^j}.$$

If $\mathcal{M} > 1$, choose a maximizing index k. Since $T_0 = 1$ and $T_1 = 0$, we have $k \ge 2$. Use (13.3) with $m = k-2$, isolate the $P_{2,k-2}T_k$ term, and divide by $P_{2,k-2}R^k\mathcal{M}$. Every other normalized neighbor is at most one by the global definition of $\mathcal{M}$; when $k = d$, the higher neighbor is $T_{d+1} = 0$ by termination. The constant, lower-order, and P_3 coefficient budgets are strict fractions whose sum is below one. Hence $|T_k|/(R^k\mathcal{M}) < 1$, contradicting maximality. We obtain:

Proposition 14.1 (Critical-point radius). *At every critical point y_0 of p_F,*

$$\max_{1 \le k \le d} \left| \frac{y_0^k p_F^{(k)}(y_0)}{p_F(y_0)} \right|^{1/k} \le K_r\sqrt{Bd}, \qquad K_r = 4096. \tag{14.2}$$

Separately, $T_0 = 1$.

The factors y_0^k and the absence of $1/k!$ in (14.2) are essential. A previous manuscript transcription deleting the former and inserting the latter was detected by numerical equation regression; the recurrence and Cauchy tail use exactly (13.4).

15 Complex Hermite–Genocchi residual

Let $E_F(z)$ be the logarithm of the exact xi/model coefficient ratio on the positive parameter branch. The six matches give

$$E_F(0) = E_F(1) = \cdots = E_F(5) = 0. \tag{15.1}$$

Six differentiations yield

$$\begin{aligned} E_F^{(6)}(z) = h^{(6)}(n+z) + \psi^{(5)}(B+z) - \psi^{(5)}(n+1/2+z) - \psi^{(5)}(A+z) \\ + \psi^{(5)}(D+z) - \psi^{(5)}(C+z). \end{aligned} \tag{15.2}$$

Pair the $B - C$ and $D - (n + 1/2)$ terms before applying the polygamma series. Let

$$\Omega = \{z \in \mathbb{C} : \mathrm{dist}(z, [0, d]) \le 2\rho\}, \qquad \rho = K_r\sqrt{Bd}. \tag{15.3}$$

The explicit wedge constant ensures

$$d + 2\rho \le \eta n \tag{15.4}$$

for a fixed small η. Therefore $n+\Omega$ stays inside the inner xi sector and every straight segment in the paired polygamma differences stays in $\Re z \ge cn$.

The absolutely convergent series

$$\psi^{(5)}(z) = 120 \sum_{k\ge 0} (z + k)^{-6}, \qquad \Re z > 0, \tag{15.5}$$

the paired mean-value estimates, and (8.4) give

$$\sup_{z\in\Omega} |E_F^{(6)}(z)| \le \frac{C_{B6}}{n^5 \log(n + 2)}. \tag{15.6}$$

Here C_{B6} is a fixed explicit constant determined by the chosen sector and parameter box. It is kept symbolic in the effectivity ledger rather than inferred from numerical samples.

We now state the complex interpolation lemma in the exact form used.

Lemma 15.1 (Six-node complex Hermite–Genocchi)**.** *Let $U \subset \mathbb{C}$ be open and convex, let f be holomorphic on U, and let $z_0, \ldots, z_5, z \in U$ with their convex hull contained in a set on which $|f^{(6)}| \le M$. If $f(z_j) = 0$ for $0 \le j \le 5$, then*

$$|f(z)| \le \frac{M}{720} \prod_{j=0}^{5} |z - z_j|. \tag{15.7}$$

Proof. Define the Newton remainders recursively. Apply the complex line-segment FTC at each stage. Six repetitions express the final divided difference as an integral of $f^{(6)}$ over the standard six-simplex. Under the stick-breaking map from $[0, 1]^6$, the Jacobian weights are

$$(1 - u_0)^5(1 - u_1)^4(1 - u_2)^3(1 - u_3)^2(1 - u_4).$$

Their nested integral is

$$\frac{1}{6}\cdot\frac{1}{5}\cdot\frac{1}{4}\cdot\frac{1}{3}\cdot\frac{1}{2}\cdot 1=\frac{1}{720}.$$

Convexity keeps every integration point in the derivative-bound set. Multiplying the divided difference by the six Newton factors gives (15.7). □

Apply the lemma to $f = E_F$ and $z_j = j$. The wedge may be enlarged so that $\rho \geq d$. Projection of $z \in \Omega$ to $[0, d]$ then gives $|z - j| \leq 3\rho$ for $0 \leq j \leq 5$. Hence

$$\begin{aligned}\sup_{z\in\Omega}|E_F(z)| &\leq \frac{C_{B6}(3\rho)^6}{720n^5\log(n+2)}\\ &\leq C_*\frac{d^3}{n^2\log(n+2)}.\end{aligned}\tag{15.8}$$

The last line uses $B \leq 3n$ and $\rho^6 = K_r^6(Bd)^3$. This is the only point at which the exponent three on d is created.

Set $c_F(z) = e^{E_F(z)}$. On the chosen branch, (15.1) implies $c_F(0) = \cdots = c_F(5) = 1$. Increasing the absolute wedge constant in (1.1) makes the right side of (15.8) small enough that

$$\sup_{z\in\Omega}|c_F(z)-1| \leq 1. \tag{15.9}$$

16 Multiplier stability and final assembly

Lemma 16.1 (Finite multiplier stability)**.** *Let $p(y) = \sum_{j=0}^{d} p_j y^j$ have d simple positive zeros and $p(0) \neq 0$. Let $\Omega_r = \{z : \operatorname{dist}(z, [0, d]) \leq 2r\}$, and suppose c is holomorphic near Ω_r, real at $0, \ldots, d$, with*

$$c(0) = \cdots = c(4) = 1, \qquad c(d) > 0, \qquad \sup_{\Omega_r}|c-1| \leq \epsilon < 16.$$

If every critical point y of p satisfies

$$\left|\frac{y^k p^{(k)}(y)}{p(y)}\right| \leq r^k, \qquad 0 \leq k \leq d, \tag{16.1}$$

then $P(y) = \sum_{j=0}^{d} p_j c(j) y^j$ has d distinct positive zeros.

Proof. Newton interpolation of the finite data gives the exact identity

$$\frac{P(y)}{p(y)} = \sum_{k=0}^{d}\frac{\Delta^k c(0)}{k!}\frac{y^k p^{(k)}(y)}{p(y)}. \tag{16.2}$$

Cauchy's estimate on discs of radius $2r$ gives $|\Delta^k c(0)|/k! \leq \epsilon/(2r)^k$. The five displayed matches make the terms $1 \leq k \leq 4$ vanish. At every critical point,

$$\left|\frac{P(y)}{p(y)} - 1\right| \leq \epsilon\sum_{k=5}^{d}2^{-k} < \frac{\epsilon}{16} < 1. \tag{16.3}$$

Thus P and p have the same nonzero sign at every critical point. They also agree in sign at zero, and $c(d) > 0$ preserves the sign for sufficiently large positive y. The simple roots of p give alternating signs at its $d-1$ critical points and the two exterior points. The intermediate-value theorem places a sign-changing zero of P in each intervening interval. Those intervals are disjoint, so the zeros are distinct. $\square$

Proof of Theorem 1.1. For $d \leq 5$, the six exact coefficient matches already identify the whole transformed polynomial with its comparison. Let $d \geq 6$. Apply Lemma 16.1 with $p = p_F$, $c = c_F$, and $r = \rho$. Simplicity and positivity come from Section 12; Proposition 14.1 gives (16.1); and (15.9) gives $\epsilon \leq 1 < 16$. Hence the transformed actual polynomial has d distinct positive zeros.

The exact coefficient normalization has the form

$$P(y) = \frac{\mathcal{J}^{d,n}(-y/S)}{\gamma(n)}, \qquad S = BR_1 > 0, \tag{16.4}$$

where R_1 is the positive first coefficient ratio on the constructed branch. The change $y = -SX$ takes distinct positive roots of P to distinct negative roots of $\mathcal{J}^{d,n}$. This proves Theorem 1.1 above the eventual analytic threshold. Put

$$K_{\text{finite}} = N_0^2(N_0 + 2) + 1.$$

If $0 \leq n < N_0$ and $d \geq 1$, then

$$n^2 \log(n+2) \leq n^2(n+1) < N_0^2(N_0+2) < K_{\text{finite}} \leq K_{\text{finite}} d^3.$$

Consequently the wedge antecedent is empty below the cutoff once $K \geq K_{\text{finite}}$. This is the finite absorption step; it invokes no unproved finite computation. Its arithmetic implication is also checked by the Lean theorem `not_twoThirdsWedge_finiteCutoffAbsorption`. $\square$

17 Effectivity and constants

The proof uses the following non-circular order of choices:

1. fix the nested sectors $\theta_0 = 1/400$, $\theta_1 = 1/200$;
2. obtain the analytic constants C_{B6} and N_{analytic} from the uniform contour and sixth-derivative proof;
3. choose $K_{\text{pre}} = 256$, then use $C_{\text{loc}} = 12 + 8\sqrt{6} < 32$;
4. set $K_0 = 256 \cdot 32^2 = 262144$ for the strengthened localization box;
5. take $C_0 = 48$, $C_1 < 96$, and $K_r = 4096$ in the recurrence;

6. enlarge the wedge constant to force every domain containment and the multiplier bound.

The exact ledger gives the deliberately coarse geometric constant

$$K_{\text{geometry}} = 52225595407351980347306803200000000000000001. \tag{17.1}$$

The residual propagation multiplies C_{B6} by an exact factor C_{mult} containing the $1/720$ simplex mass, the six Newton factors, $B \leq 3n$, K_r^6, and the exponential conversion from E_F to $c_F - 1$. A sufficient final constant is

$$K_{\text{final}} = \max\left\{K_{\text{geometry}},\ C_{\text{mult}}C_{B6}, K_{\text{finite}}\right\}, \quad N_0 = \max\{N_{\text{analytic}}, N_{\text{explicit}}\}. \tag{17.2}$$

No empirical diagnostic enters (17.2). The often useful numerical observation that a remainder resembles $10.7/L_n$ is excluded from the theorem ledger.

18 Formal and computational verification

The accompanying Lean development distinguishes four layers.

1. *Kernel-checked concrete analytic chain*: the exact xi/theta/Mellin identity and factor-eight normalization, the quantitative fixed-sector saddle branch, legal leading contour and Gaussian localization, uniform infinite higher-mode suppression, fixed-sector Gamma/Stirling assembly, and the holomorphic coefficient theorem through six derivatives.

2. *Kernel-checked xi specialization*: the moving-saddle derivative tower, elementary cube and whole-box estimates, exact positive parameter branch, quotient-to-six-coefficient adapter, terminating hypergeometric producer and ODE, actual transformed xi polynomial, and six matches. With typed classical root inputs, Lean further checks the finite-free specialization, root and critical-point localization, recurrence budgets, the literal normalized critical-radius bound, the concrete analytic multiplier and its six nodes, the finite Newton–Cauchy estimate, the canonical Rolle-point interval certificate, and the actual Riemann-xi Jensen headline theorem. The terminal declarations additionally prove that the Jensen polynomial has degree exactly d, rule out $d+1$ distinct negative zeros, and combine the analytic and pre-cutoff cases into the single global declaration `riemannXiJensen_twoThirds_global_headline_exactly`.

3. *Imported formal boundary*: the general Jacobi matrix/root theorems, MMP log-mesh preservation, and MSS product-root inequality are imported as narrow typed records rather than re-proved. They are the only mathematical hypotheses of the concrete headline Lean theorem beyond its displayed cutoff and wedge conditions. The factor-root and degree fields in the MMP and MSS records are explicit domain side conditions

on the imported results; because the general literature theorems are not re-proved, those fields document and type-check applicability rather than deriving the imported conclusions internally.

4. *Computational corroboration*: exact CAS identities, directed Arb/ACB enclosures, interval ledgers, and finite numerical root examples.

There is no Lean axiom asserting the xi asymptotic, an imported root theorem, or the headline result. Literature inputs are structure fields supplied to theorems, not axioms added to the environment. The Phase-26 audit covers 1,217 declarations and reports only Lean's standard logical principles `propext`, `Classical.choice`, and `Quot.sound`; the source scan finds no `sorry`, `admit`, custom `axiom`, or unsafe escape. The generic end-to-end theorem remains available through typed certificate interfaces. The later Phase-30 terminal theorem constructs the xi-specific multiplier certificate and proves the concrete Riemann-xi Jensen root conclusion, while continuing to expose the Jacobi/MMP/MSS literature records as hypotheses rather than axioms.

Phase 28 adds a terminal axiom audit for the three literal paper-facing T5 declarations: the effective and existential forms of Theorem 7.1, and its relative-error derivatives through order six. Phase 29 packages those declarations as a short Mathlib-only Palomar Challenge/Solution topic and passes the local source-fidelity, semantic mutation, fresh-kernel, and archive checks. This is a submission candidate, not an official Palomar result: Comparator, NanoDa, and editorial replay remain pending until public submission.

The post-Phase-32 repair audit additionally checks the guarded MSS interface, both strict lower-endpoint proofs, the exact degree/root-count adapter, and the single global cutoff assembly. A multiline axiom parser checks every name in each `#print axioms` block and rejects a synthetic custom axiom placed on a continuation line. A regression attack that reproduces the former unguarded negative-endpoint call is required to fail elaboration at the new positivity argument.

The clean-room Mathematica 15.0.1 notebook reads no repository artifact as mathematical input. It independently reconstructs the saddle derivatives, the 82-term H_6 numerator, the leading system/Jacobian, and the shifted hypergeometric recurrence. The canonical result-ledger SHA-256 is

`1faea5fcb35b504b5d0aad9391999a99e530373dce36addb496eabb133fb04cc`.

The notebook was executed by the author using cells prepared with AI assistance. It is evidence of independent CAS evaluation, not independent human review.

Arb/ACB directly encloses the first centered xi coefficients, two independently normalized integral paths with analytic tails, three small Rouché boxes, the implicit derivative tower, and selected finite Jensen root boxes. Finite contour grids are labeled regression evidence and are not used to infer a uniform theorem.

19 Trust boundary and availability

This manuscript has not received human mathematical review or peer review. The separated analytic, algebraic, Lean, and reproducibility reviews in the release workflow are AI adversarial reviews. They are reported as such. The theorem is presented with a complete paper proof, but the absence of human review is a material limitation for circulation of a new result.

The reviewer package contains the main paper, technical supplement, synopsis, public explainer, Lean sources and lockfiles, Mathematica notebook/PDF/ledger, exact SymPy and interval calculations, Arb/ACB scripts and results, source hashes, verification commands, and machine-readable assurance/dependency ledgers. None of these artifacts should be described as a substitute for scholarly judgment. The package is designed so a reviewer can begin with the theorem-to-evidence cross-reference and then reproduce any channel without relying on another channel's saved numerical answer.

The public-release set is versioned independently of the private development repository. Its version-specific archive, checksums, source commit, and permanent record should be cited together. The Version 1.1 DOI is `10.5281/zenodo.22293642`. Later corrections are to be issued as new numbered versions; the files of this version are not to be silently replaced.

For Theorem 7.1, the repository additionally contains a locally verified Palomar publication candidate. Its existence does not mean that Palomar has accepted or replayed it. If an official result is later available, the public entry and immutable identifier must be added here before any Palomar verification claim is made.

Declaration of AI and AI-assisted technologies

These papers and the associated Lean 4 formalization were developed with substantial artificial-intelligence assistance, primarily from **OpenAI GPT-5.6 Sol Pro**. Additional models assisted with adversarial testing, source comparison, independent recalculation, and AI-only technical review. **Wolfram Mathematica 15.0.1** was used in a clean-room computation executed by the author to check selected symbolic identities and exact algebraic calculations. Those computations provide independent-CAS corroboration of the specified calculations; they do not constitute a proof of the entire theorem, human mathematical review, or peer review. The author directed, reviewed, and edited the work, authorized the repository changes, and accepts responsibility for the contents. AI systems are not authors.

The model-specific review reports identify their provider, scope, context, and limitations.

A Theorem-to-evidence map

T1: xi/Mellin identity. Paper derivation and unconditional Lean proof of the concrete theta/Mellin/omega bridge and factor-eight coefficient identity; exact CAS and Arb corroboration.

T2: saddle branch. Lemma 5.1; Lean whole-disc contraction, unique fixed-sector branch, holomorphy, curvature, and explicit boxes; Arb boxes provide a separate finite check.

T3–T5: contour and coefficient theorem. Sections 6–7; Theorem 7.1 is followed by its proof, with the estimates expanded in Appendix D. Lean checks the legal contour, Gaussian localization, full higher-mode sum, Gamma/Stirling assembly, the literal 1/100–1/200–1/400 sector geometry of Theorem 7.1, and its order-six Cauchy consequence. Directed regressions remain corroboration rather than proof premises.

T6: six derivatives.

Exact rational tower; Lean Cauchy transport; SymPy/Mathematica/ACB agreement.

T7–T9: six matches and branch.

Lean checks the quotient adapter, cube calculus, exact xi whole-box branch, actual transformed polynomial, and six coefficient matches at an explicit cutoff.

T10–T12: finite-free roots. Lean checks every convention, scaling, Gershgorin, reciprocal, product, and localization adapter; classical Jacobi facts, MMP v3, and MSS Theorem 1.6 remain typed external inputs.

T13–T14: recurrence and radius. Lean checks the finite producer, ODE, complete maximum argument, explicit coefficient budgets, critical-point localization, and literal xi comparison radius under the typed root inputs.

T15–T18: residual and assembly. Lean checks the local complex HG and finite Newton identities, constructs the actual xi multiplier and its uniform unit bound, builds the complete Rolle-point interval certificate, performs sign transfer, identifies the transformed Jensen polynomial, and proves the concrete headline conclusion conditional only on the typed Jacobi/MMP/MSS literature inputs. The explicit paper proof of Theorem 1.1 closes Section 16.

B Normalization and domain audit

This appendix expands the interfaces that are especially vulnerable to a factor, sign, or domain error. It is part of the proof, rather than a report about the accompanying computations. The technical supplement records the machine implementations of the same interfaces.

B.1 The entire completed function

Let Λ_0 denote the pole-removed entire completion used in the formal library. Its relation to the meromorphic completion is

$$\Lambda(s) = \Lambda_0(s) - \frac{1}{s} - \frac{1}{1-s}.$$

The entire function in this paper is therefore

$$\xi(s) = \frac{s(s-1)}{2}\Lambda_0(s) + \frac{1}{2}. \tag{B.1}$$

For $s \neq 0, 1$, substitution of the preceding identity in (B.1) cancels the two polar terms and gives the usual $s(s-1)\Lambda(s)/2$. Both sides are entire, so the equality extends across 0 and 1. The functional equation $\Lambda_0(1-s) = \Lambda_0(s)$ then gives

$$\xi(1-s) = \xi(s).$$

Consequently $w \mapsto \xi(1/2 + w)$ is even and its odd derivatives at zero vanish.

The coefficient convention is worth checking at the derivative level. From

$$\xi(1/2 + w) = \sum_{n \geq 0} \frac{\gamma(n)}{n!} w^{2n}$$

one obtains exactly

$$\gamma(n) = \frac{n!}{(2n)!} \left. \frac{\mathrm{d}^{2n}}{\mathrm{d}w^{2n}} \xi(1/2 + w) \right|_{w=0}. \tag{B.2}$$

There is no factor 2^{2n} in (B.2). This explicit derivative check is also an equation-regression gate in the release package.

B.2 The factor-eight calculation

Riemann's theta-kernel identity, after the substitution $t = e^{2u}$ and symmetrization about the critical line, gives

$$\left. \frac{\mathrm{d}^{2n}}{\mathrm{d}w^{2n}} \xi(1/2 + w) \right|_{w=0} = 8 \int_0^\infty \omega(e^{2u}) e^{u/2} u^{2n} \, \mathrm{d}u. \tag{B.3}$$

Combining (B.2) and (B.3) gives the moment formula used in Section 4. The factor $e^{u/2}$ is the Jacobian and centered power contribution after $t = e^{2u}$; replacing it by e^u would change even the zeroth moment. The release regression evaluates the two sides independently at low orders precisely to detect either transcription error.

For the later saddle calculation it is convenient to integrate by parts and write

$$F(s) = \int_0^\infty \mathcal{K}(t) t^s \, \mathrm{d}t.$$

The boundary terms vanish because the theta kernel decays exponentially at infinity and its modular transform supplies the corresponding decay at zero. Two integrations give

$$\gamma(M) = \frac{\Gamma(M+1)}{\Gamma(2M+1)} 2^{-2M-2} \left(32 \binom{2M}{2} F(2M-2) - F(2M) \right). \tag{B.4}$$

The two Mellin arguments in (B.4) explain the shift $N = 2M-2$ used throughout the saddle analysis. Define

$$M_z = 2^{-2z-2} \left(32 \binom{2z}{2} F(2z-2) - F(2z) \right),$$

so that $\gamma(z) = \Gamma(z+1)M_z/\Gamma(2z+1)$ exactly.

B.3 Nested sectors and proportional discs

Fix

$$\mathcal{S}_1(R) = \{M : |M| > R,\ |\arg M| < 1/200\}, \qquad \overline{\mathcal{S}}_0(R) = \{M : |M| \geq R,\ |\arg M| \leq 1/400\}.$$

All logarithms and square roots are continued from the positive axis in $\mathcal{S}_1(R)$. There is an $\eta_0 > 0$, depending only on the difference between the two displayed angles, such that

$$M \in \overline{\mathcal{S}}_0(R), \quad |z - M| \leq \eta_0 |M| \quad \Longrightarrow \quad z \in \mathcal{S}_1(R/2). \tag{B.5}$$

Indeed, $|z| \geq (1-\eta_0)|M|$, while the angular displacement is at most $\arcsin(\eta_0/(1-\eta_0))$. Choose η_0 so this is below $1/400$. This elementary separation is the geometric input that permits Cauchy differentiation of the relative error. It also explains why the main-term estimate may be stated on a wider sector than the final uniform error.

C Detailed saddle estimates

C.1 Legality of the logarithmic equation

Put $\ell = \log |N|$ and

$$L_0 = \log N - \log\log N - \log \pi.$$

On the closed disc $D_N = \{L : |L - L_0| \leq 1\}$ define

$$m(\ell) = \ell - \log(\ell+1) - \log \pi - 1,$$
$$M_\theta(\ell) = \ell + 2 + \log(\ell+1) + \theta/\ell + \log \pi.$$

For $\ell \geq 12$ and $|\arg N| \leq \theta$ the triangle inequality gives

$$\Re L \geq m(\ell) > 0, \qquad |L| \leq M_\theta(\ell). \tag{C.1}$$

Thus the principal logarithm of L is legal on the whole disc. Moreover,

$$\left|\frac{3L}{4N}\right| \leq \frac{3M_\theta(\ell)}{4e^\ell} < 10^{-4}, \tag{C.2}$$

so the correction logarithm is legal there as well.

For $|z| \leq 1/2$, integration along the segment from 0 to z gives

$$\log(1+z) = \int_0^1 \frac{z}{1+tz}\,\mathrm{d}t, \qquad |\log(1+z)| \leq 2|z|.$$

Apply this first to $L/L_0 - 1$ and then to $-3L/(4N)$. Since $|L_0| \geq \ell/2$, the comparison function in the proof of Lemma 5.1 satisfies

$$\sup_{\partial D_N} |G_N(L) - (L - L_0)| \leq 2\frac{\log(\ell+1) + \theta/\ell + \log\pi + 1}{\ell} + \frac{3M_\theta(\ell)}{2e^\ell}. \tag{C.3}$$

The right side is decreasing from $\ell = 12$ and is strictly below one at that endpoint. Rouché therefore gives one zero counted with multiplicity. No numerical sample of an interior zero is used in this conclusion.

C.2 Simplicity before differentiation

At the zero, exponentiation of the logarithmic equation yields $N = L_N(\pi e^{L_N} + 3/4)$. Set $r = L_N^{-1}$ and $\sigma = L_N/N$. The derivative denominator factors as

$$Q_N = (1 + L_N)N - \frac{3}{4}L_N^2 = NL_N\left(1 + r - \frac{3}{4}\sigma\right). \tag{C.4}$$

After enlarging R, the disc geometry gives $|r|, |\sigma| \leq 7/50$, whence

$$\left|1 + r - \frac{3}{4}\sigma\right| \geq 1 - \frac{7}{50} - \frac{21}{200} = \frac{151}{200} > \frac{9}{16}. \tag{C.5}$$

Thus the root is simple before the implicit-function theorem is invoked. Local branches patch because any two branches meeting the same Rouché disc must select its unique zero. Differentiation now legitimately gives $L_N' = L_N/Q_N$.

The denominator arising after six differentiations is a power of $4 + 4r - 3\sigma$. The same box gives the stronger margin

$$|4 + 4r - 3\sigma| \geq 4 - \frac{28}{50} - \frac{21}{50} = \frac{151}{50}. \tag{C.6}$$

Equations (C.5) and (C.6) are kept separate: the first licenses analytic differentiation; the second controls the exact rational derivative tower.

C.3 Imaginary part and Gaussian direction

Write $L_N = a + ib$ and $N = |N|e^{i\phi}$. Taking arguments in the saddle equation gives

$$\phi = \arg L_N + \arg(\pi e^{L_N} + 3/4).$$

Because $a \geq m(12) > 7.29$, the second argument differs from b by at most $3e^{-a}/(2\pi)$, and $|\arg L_N| \leq |b|/a$. Therefore

$$|b| \leq |\phi| + \frac{|b|}{a} + \frac{3e^{-a}}{2\pi}. \tag{C.7}$$

Solving (C.7) at $|\phi| \leq 1/100$ gives $|b| < 0.012 < 1/20$. This direct argument avoids deriving the imaginary bound from the much coarser estimate $|L_N - L_0| < 1$.

Finally,

$$K_N = \frac{N}{L_N}\left(1 + \frac{1}{L_N} - \frac{3L_N}{4N}\right).$$

The last factor is in the disc of radius 49/200 about 1. Combining its argument with (C.7), and increasing R once more, gives

$$|\arg K_N| < 1/20, \qquad \Re K_N \geq \cos(1/20)|K_N|. \tag{C.8}$$

The right-half-plane inequality, not merely $K_N \neq 0$, supplies the uniform real Gaussian majorant.

D Contour and coefficient details

D.1 Truncation and connectors

Set

$$\Phi_N(u) = N \operatorname{Log} u - \frac{3}{4}u - \pi e^u, \qquad g_N(u) = e^u e^{\Phi_N(u)}, \qquad I_1(N) := F_1(N) := \int_0^\infty g_N(u)\,\mathrm{d}u.$$

Write $L_N = \alpha + ib$ and first split this integral at 1. Apply Cauchy's theorem to the finite rectangle with vertices $1, X, X + ib, 1 + ib$. The right connector tends to zero as $X \to \infty$, so

$$I_1(N) = E_N + \int_1^\infty g_N(x + ib)\,\mathrm{d}x, \qquad E_N = \int_0^1 g_N(x)\,\mathrm{d}x + i\int_0^b g_N(1 + iy)\,\mathrm{d}y. \tag{D.1}$$

The horizontal ray lies in $\Re u \geq 1$, hence in a fixed principal-log domain. The finite endpoint term satisfies

$$|E_N| \leq |\mathcal{A}(N)|e^{-c'|N|\log\log|N|}. \tag{D.2}$$

This order of limits prevents the invalid shortcut of translating the whole infinite real line across the logarithmic cut.

D.2 Central window and odd cancellation

On the actual shifted ray write $u = L_N + r$, so $r \geq 1 - \Re L_N$, and choose $V = |K_N|^{-2/5}$. For sufficiently large $|N|$, the central window is contained in this ray. Taylor's formula with integral remainder yields, uniformly for $|r| \leq V$,

$$\Phi_N(L_N + r) - \Phi_N(L_N) = -\frac{1}{2}K_N r^2 + \frac{1}{6}\kappa_3 r^3 + R_4(N, r), \tag{D.3}$$

where

$$|R_4(N, r)| \leq C|K_N||r|^4. \tag{D.4}$$

The Jacobian outside the phase contributes e^r. By (C.8), $e^{-K_N r^2/2}$ is dominated by $e^{-c|K_N|r^2}$. Extending the central comparison to the full real Gaussian introduces only an exponentially small left tail; the full real Gaussian is not the deformed contour. Completing the square gives

$$\frac{\int_{\mathbb{R}} r^3 e^{r - K_N r^2/2}\, \mathrm{d}r}{\int_{\mathbb{R}} e^{r - K_N r^2/2}\, \mathrm{d}r} = \frac{3}{K_N^2} + \frac{1}{K_N^3}.$$

Since $\kappa_3 = O(|K_N|)$, the signed cubic, its square, and the quartic remainder contribute $O(|K_N|^{-1})$ relatively. Estimating the absolute value of the cubic before integration would lose this order.

Outside the central window on the actual range $r \geq 1 - \Re L_N$, strict horizontal concavity makes the left tail increase up to the saddle window and gives a negative derivative of size $\gg |K_N|^{3/5}$ at the right edge. Splitting at $|r| = 1$ reduces the two resulting tail bounds to elementary Gaussian and exponential integrals. Together with (D.2), this proves the leading-mode formula (6.3).

D.3 Modes and relative error

The theta mode $m \geq 2$ differs from the leading mode by $-\pi(m^2 - 1)e^{L_N + r}$. On the central window, (C.7) keeps its real part negative, while the saddle equation gives $e^{\Re L_N} \asymp |N|/\log|N|$. Thus

$$|F_m(N)| \leq |\mathcal{A}(N)| \exp\left(-c(m^2 - 1)|N|/\log|N|\right). \tag{D.5}$$

The series in m is dominated by its first term. This bound is uniform on the fixed sector; it is not asserted with a constant uniform as the sector angle approaches $\pi/2$.

For $a(s) = \log \mathcal{A}(s)$, use the saddle identity during differentiation to obtain

$$a'(s) = \log L_s + (L_s K_s)^{-1} - \tfrac{1}{2}K_s'/K_s, \qquad a''(s) = O((|s|\log|s|)^{-1}).$$

Taylor's formula on a segment of length two gives

$$\frac{F(N+2)}{F(N)} = L_N^2\left(1 + O\left(\frac{\log|N|}{|N|}\right)\right).$$

Combining this with (B.4), the leading-mode formula, and sectorial Stirling proves Theorem 7.1. Holomorphy of the relative error follows because the main term is nonzero on the outer sector and every preceding estimate was established there before restriction to the inner sector.

E Derivative and polygamma audit

After Theorem 7.1, M_z is nonzero on a sufficiently remote outer sector. Throughout this section

$$h(z) = \log M_z,$$

on the branch real on the positive axis. In particular, h is not $\log \gamma(z)$; the gamma-ratio derivatives in the exact identity above are kept separate.

E.1 Implicit derivative coordinates

The direct differentiation is organized in the two small variables $r = L_N^{-1}$ and $\sigma = L_N/N$. From (C.4),

$$L'_N = \frac{1}{N(1 + r - 3\sigma/4)}. \tag{E.1}$$

Every subsequent derivative of r and σ is obtained from (E.1) by the product and quotient rules. This keeps the denominator visible: through order six it is a power of $4 + 4r - 3\sigma$, whose nonvanishing has already been proved in (C.6).

There are two differentiation variables in the final coefficient problem. The saddle main term is

$$G_0(N) = (N + 1) \log L_N + \frac{L_N}{4} - \frac{N}{L_N} - \frac{1}{2} \log Q_N.$$

First differentiate G_0 with respect to N. Then apply $N = 2x - 2$. The chain contributes a factor 2^j at derivative order j, while the prefactor powers of N and L_N absorb the corresponding powers in the reduced normalization. The resulting leading constants at orders $2, 3, 4, 5, 6$ are

$$2, -2, 4, -12, 48.$$

At orders five and six the constants before the final chain are -6 and 24. Recording both lists is a guard against applying the factor two only once rather than at every differentiation.

At fifth order, setting $\sigma = 0$ in the exact reduced function gives

$$-\frac{6 + 47r + 146r^2 + 210r^3 + 126r^4 + 42r^5 + 6r^6}{(1 + r)^7}. \tag{E.2}$$

At sixth order the numerator has 82 nonzero monomials and total degree 13, over $(4 + 4r - 3\sigma)^{12}$. The canonical expanded numerator is kept in the exact computation ledger rather than typeset twice. A coefficientwise triangle inequality on $|r|, |\sigma| \leq 7/50$ gives exactly

$$\frac{64221398057649315840365335551104}{70257609972813759418868 4005} < 10^4. \tag{E.3}$$

Both sides of (E.3) are rational. No rounded decimal is used to prove the strict inequality.

E.2 Cauchy transport of the relative error

Write $\gamma = \mathcal{G}(1+\mathcal{E})$ on the outer sector. For a positive x in the inner sector choose $R_x = \eta_0 x$, with η_0 from (B.5). Cauchy's formula gives

$$|\mathcal{E}^{(j)}(x)| \leq \frac{j!}{(\eta_0 x)^j} \sup_{|z-x|=\eta_0 x} |\mathcal{E}(z)|.$$

The sectorial bound on the circle is $O(\log x/x)$. Leibniz expansion of $\log(1+\mathcal{E})$ is legal after increasing the threshold so that $|\mathcal{E}| \leq 1/2$, and every derivative through order six is lower order than the corresponding term of the saddle tower. This is the step that turns a holomorphic relative coefficient asymptotic into the uniform logarithmic derivative estimate; differentiating a real-axis big-O term would not be valid.

E.3 Paired polygamma differences

For $\Re z > 0$,

$$\psi^{(5)}(z) = 120 \sum_{k=0}^{\infty} (z+k)^{-6}.$$

The terms are absolutely and locally uniformly convergent, so they may be differentiated and paired on the thickened domain. The $B - C$ contribution is treated as one difference:

$$\psi^{(5)}(B+z) - \psi^{(5)}(C+z) = (B-C) \int_0^1 \psi^{(6)}(C+z+t(B-C))\, \mathrm{d}t.$$

The $D - (n+1/2)$ pair is handled identically. Domain containment keeps every segment in $\Re z \geq cn$, and the series bounds the sixth polygamma by a constant times n^{-6}. The lengths of the paired segments carry the small branch differences that would be lost by estimating the two polygamma terms separately. Combining these paired estimates with the transported xi derivative proves (15.6).

F Cube calculus and the limiting branch

F.1 A direct proof of the cube identity

Let $f_0(U) = \log U - \log(U+1)$. For $q = j+1$, apply the real fundamental theorem of calculus successively to each forward difference. Since

$$\frac{\mathrm{d}^q}{\mathrm{d}U^q} \log U = (-1)^{q-1}(q-1)!U^{-q},$$

Fubini gives

$$\Delta^j f_0(U) = (-1)^{j+1} j! \int_{[0,1]^q} (U + u_1 + \cdots + u_q)^{-q}\, \mathrm{d}u.$$

The integrand is continuous and positive on the parameter box, so no improper integration or interchange issue occurs.

For

$$\Phi_q(s,z) = \int_{[0,1]^q} (s + z(u_1 + \cdots + u_q))^{-q}\, \mathrm{d}u,$$

differentiate under the integral. The first two parameter derivatives are

$$\partial_s \Phi_q(s,z) = -q \int_{[0,1]^q} (s + z\Sigma u)^{-q-1}\, \mathrm{d}u,$$
$$\partial_s^2 \Phi_q(s,z) = q(q+1) \int_{[0,1]^q} (s + z\Sigma u)^{-q-2}\, \mathrm{d}u.$$

If $s \ge s_0 > 0$ and $z \ge 0$, the unit cube has mass one and therefore

$$|\partial_s^r \Phi_q(s,z)| \le (q)_r s_0^{-q-r}, \qquad r = 0, 1, 2. \tag{F.1}$$

Apply the mean-value theorem to the z variable. Since $0 \le \Sigma u_i \le q$,

$$\left|\partial_s^r \Phi_q(s,z) - (-1)^r (q)_r s^{-q-r}\right| \le (q)_r (q+r) q z\, s_0^{-q-r-1}. \tag{F.2}$$

These two inequalities are the entire analytic content of the elementary cube stage.

F.2 Three limiting mechanisms

Substitute (F.1)–(F.2) in (10.6). The remote A endpoint carries the external factor e^{q-1}: it survives only for $q = 1$. The $B - C$ pair is a divided difference across a segment of length we and tends to qw/t^{q+1}. The D term must be paired with the gamma half-shift before rescaling; its segment length tends to $\delta e - x/2$ and its normalized limit is $q\delta$. Keeping these mechanisms separate gives exactly

$$H^\infty(\alpha, t, w, \delta) = \left(\alpha^{-1} + wt^{-2} + \delta, wt^{-3} + \delta, 3wt^{-4} + 3\delta, 4wt^{-5} + 4\delta\right).$$

The value error is $O(e + x/e)$ and the first-derivative error is $O(e + x)$, uniformly on the rational box. The estimates are obtained from compact endpoint inequalities, not from a plot of the limiting branch.

F.3 Exact limiting solution and inverse

Adding $S^\infty = (-2, -1, -2, -2)$ gives residuals $F_0, \ldots, F_3$. Direct subtraction yields

$$3F_1 - F_2 = \frac{3w(t-1)}{t^4} - 1, \qquad \frac{4}{3}F_2 - F_3 = \frac{4w(t-1)}{t^5} - \frac{2}{3}.$$

At a zero, these give $3w(t-1) = t^4$ and $6w(t-1) = t^5$; positivity forces $t = 2$, then $w = 16/3$, $\delta = 1/3$, and the first equation gives $\alpha = 3$. Thus the positive solution is $y_* = (3, 2, 16/3, 1/3)$.

The exact inverse of the Jacobian in (11.3) is

$$P = \begin{pmatrix} -9 & 45 & -24 & 9 \\ 0 & 6 & -6 & 3 \\ 0 & 48 & -112/3 & 16 \\ 0 & 1 & -4/3 & 1 \end{pmatrix}. \tag{F.3}$$

Direct multiplication in both orders gives the identity matrix, and the largest absolute row sum is 304/3. For $T_n(y) = y - PG_n(y)$, the two required certificates are

$$\|G_n(y_*)\|_\infty \le \frac{3R}{608}, \qquad \sup_{\|y-y_*\|_\infty \le R} \|DT_n(y)\|_\infty \le \frac{1}{2}. \tag{F.4}$$

The first inequality and (F.3) give $\|T_n(y_*) - y_*\|_\infty \le R/2$; the second supplies both the self-map and the contraction. Checking the derivative only at the center would not be sufficient.

The outer rational box has respective left and right margins

$$(1/2, 1/2), \quad (1/4, 1/4), \quad (1/3, 2/3), \quad (1/12, 1/12)$$

about y_*. For $e \le 1/12$, endpoint arithmetic proves the ordering inequalities (11.8). Thus every denominator used in the finite model is positive on the whole certified branch box.

G Finite-free orientation and root localization

G.1 The convention adapter

Write ascending constant-term-one polynomials as

$$p(X) = \sum_{k=0}^{d} (-1)^k a_k X^k, \qquad q(X) = \sum_{k=0}^{d} (-1)^k b_k X^k.$$

Their multiplicative finite-free convolution is

$$(p \boxtimes_d q)(X) = \sum_{k=0}^{d} (-1)^k \binom{d}{k}^{-1} a_k b_k X^k. \tag{G.1}$$

The cited literature states its theorems for monic descending polynomials. Let $\operatorname{rev}_d p = X^d p(1/X)/p(0)$. Reindexing the finite sum in (G.1) gives

$$\operatorname{rev}_d(p \boxtimes_d q) = \operatorname{rev}_d(p) \boxtimes_d \operatorname{rev}_d(q).$$

If $p(0) \ne 0$, reversal maps every positive root to its reciprocal. Hence an interval $[a, b]$ is transported to $[b^{-1}, a^{-1}]$, and strict logarithmic mesh is preserved. This proves that the orientation change does not silently reverse either a root bound or a mesh inequality.

G.2 Explicit MMP specialization to the comparison ${}_3F_2$

This subsection spells out the specialization that is easy to obscure in a generic "finite-free input." For $U, V, s > 0$ define

$$Q^{(s)}_{U,V}(X) = {}_2F_1\left(\begin{matrix}-d, U\\ V\end{matrix}; \frac{sX}{U}\right) = \sum_{k=0}^{d}(-1)^k \binom{d}{k}\frac{(U)_k}{(V)_k}\left(\frac{s}{U}\right)^k X^k. \tag{G.2}$$

Thus, in the notation of (G.1), the unsigned coefficient products of the two factors $Q^{(1)}_{A,B}$ and $Q^{(D)}_{C,D}$ are

$$a_k = \binom{d}{k}\frac{(A)_k}{(B)_k}A^{-k}, \qquad b_k = \binom{d}{k}\frac{(C)_k}{(D)_k}\left(\frac{D}{C}\right)^k.$$

Substitution in (G.1) gives

$$\begin{aligned}
[Q^{(1)}_{A,B} \boxtimes_d Q^{(D)}_{C,D}](X) &= \sum_{k=0}^{d}(-1)^k \binom{d}{k}\frac{(A)_k(C)_k}{(B)_k(D)_k}\left(\frac{D}{AC}\right)^k X^k \\
&= {}_3F_2\left(\begin{matrix}-d, A, C\\ B, D\end{matrix}; \frac{D}{AC}X\right) = p_F(X).
\end{aligned} \tag{G.3}$$

This is an identity of finite coefficient lists, with no root theorem used. The Lean theorem `terminating3F2Polynomial_eq_finiteFree_jacobiFactors` proves the same identity for symbolic positive A, B, C, D, and `xiNaturalComparisonPolynomial_eq_finiteFree` substitutes the actual xi branch parameters.

It remains to check that the two factors in (G.3) are in MMP's positive-rooted class. The standard Jacobi identity is

$$Q^{(s)}_{U,V}(X) = \frac{d!}{(V)_d}P_d^{(V-1,\,U-V-d)}\left(1 - \frac{2sX}{U}\right). \tag{G.4}$$

On the branch used in the proof,

$$B, D > 0, \qquad A \geq B + d, \qquad C \geq D + d, \qquad 1, D > 0.$$

For either factor in (G.4), the Jacobi parameters $V - 1$ and $U - V - d$ are therefore greater than -1. The classical Jacobi zero theorem gives d simple zeros in $(-1, 1)$. The affine map in (G.4) sends them to

$$X_j = \frac{U}{2s}(1 - x_j) > 0.$$

The top coefficient in (G.2) is nonzero, so each factor has degree exactly d; its constant term is one. These facts verify membership in the positive-rooted degree-d class required by MMP, not merely numerical plausibility for selected parameters.

Now reflect both factors to MMP's monic descending convention. The reflection identity proved above shows that their MMP convolution is precisely the reflection of the left side of (G.3). MMP v3 Proposition 2.7(iii) places all convolution roots in $[0, \infty)$. Since the ascending constant term is one, zero is excluded. For $d \geq 2$, the simple positive roots of the first Jacobi factor have

$$\mathrm{lmesh}(Q^{(1)}_{A,B}) > 1.$$

MMP v3 Definition 2.16 and Proposition 2.17 give

$$\mathrm{lmesh}(Q^{(1)}_{A,B} \boxtimes_d Q^{(D)}_{C,D}) \geq \mathrm{lmesh}(Q^{(1)}_{A,B}) > 1,$$

so the product roots are distinct. For $d = 0$ the assertion is vacuous, and for $d = 1$ the nonzero linear coefficient gives one simple positive root. Equation (G.3) finally transfers the conclusion to the paper's particular ${}_3F_2$.

The formal boundary now mirrors exactly this calculation. `MMPFiniteFreeLogMeshInput` takes the two displayed factor polynomials, their complete positive-root and degree certificates, and the MMP conclusion for their finite-free convolution. It does not take the final xi comparison function as an assumed positive-rooted object. Lean then rewrites the convolution to that function using the independently proved coefficient identity. The general Jacobi zero theorem and the two general MMP propositions remain declared literature inputs; the entire specialization from those inputs to this ${}_3F_2$ is explicit.

The MSS boundary is guarded in the same way. Its record includes the two positive-root and degree certificates and accepts a product interval only after proofs of $0 < u_-$ and $0 < v_-$. At the concrete call site, Lean derives

$$u_- = B - 8\sqrt{Bd} > 0, \qquad v_- = 1 - 8\sqrt{d/D} > 0$$

from $B, D \geq 256d$. This guard is essential: allowing arbitrary negative lower endpoints would make the universally quantified interval record inconsistent as soon as the convolution had a real root. The factor fields in the MMP and MSS records are domain side conditions for externally supplied literature conclusions; they are deliberately visible even though the general literature proofs themselves are outside the Lean development.

G.3 A ratio-free Jacobi interval

Transport the classical Jacobi matrix from $[-1, 1]$ to the positive-root coordinate. Under $U \geq V + d$ and $V \geq 32d$, its diagonal displacement and adjacent off-diagonal row sum satisfy

$$|J_{ii} - V| \leq 4\sqrt{Vd}, \qquad |J_{i,i-1}| + |J_{i,i+1}| \leq 4\sqrt{Vd}.$$

Gershgorin's theorem therefore places every root in

$$[V - 8\sqrt{Vd}, V + 8\sqrt{Vd}]. \tag{G.5}$$

This estimate does not assume that the ratio of the two finite model gaps is small. That distinction matters because the new branch has $(C - D)/D \to 1$, whereas the sharper estimate used in the earlier argument requires this ratio to be at most $1/4$.

The preliminary inequalities in (12.6) make both Jacobi intervals positive. In particular the safe threshold is $K_{\mathrm{pre}} = 256$: using 32 in the generic second factor would give the lower relative endpoint $1 - 8/\sqrt{32} < 0$. Applying the maximum-root product theorem in the original and reciprocal orientations gives both endpoints of the convolution interval. If the deviations of the two factors are u and v, then

$$\begin{aligned}|(B+u)(1+v) - B| &\le 8\sqrt{Bd} + 8B\sqrt{d/D} + 64\sqrt{Bd}\sqrt{d/D} \\ &\le (12 + 8\sqrt{6})\sqrt{Bd}.\end{aligned}$$

This is $C_{\mathrm{loc}} = 12 + 8\sqrt{6}$, not the transposed expression $8 + 12\sqrt{6}$.

H Recurrence and global-maximum argument

H.1 Finite coefficient proof

For fixed m, initialize $c_0 = 1$ and impose the exact finite recurrence

$$AC(k+1)(k+B+m)(k+D+m)c_{k+1} = D(k+m-d)(k+A+m)(k+C+m)c_k. \quad \text{(H.1)}$$

At $k = d - m$ the right side vanishes, so the producer terminates. Every denominator on the preceding range is positive. Equation (H.1) therefore proves the coefficient ratio without asking a symbolic system to simplify a quotient of analytically continued Pochhammer functions at unspecified arguments.

On a monomial y^k, the Euler operator $\vartheta = y\,\mathrm{d}/\,\mathrm{d}y$ acts by multiplication by k. Applying both sides of (13.2) to the finite coefficient list and using (H.1) proves the ODE coefficient by coefficient. The shift identity $(a)_{k+m} = (a)_m(a+m)_k$ identifies the shifted polynomial with the mth derivative up to a nonzero factor.

Expanding the ODE gives the equivalent coefficients

$$\begin{aligned}
P_{3,m} &= \frac{AC - Dy}{AC}, \\
P_{2,m} &= B + D + 2m + 1 - \frac{Dy}{AC}(A + C + 3m - d + 3), \\
P_{1,m} &= (B+m)(D+m) - \frac{Dy}{AC}\big((A+m)(C+m) + (m-d)(A+C+2m+1)\big), \qquad \text{(H.2)} \\
P_{0,m} &= -\frac{Dy}{AC}(m-d)(A+m)(C+m).
\end{aligned}$$

The final coefficient is nonnegative for $0 \le m \le d$ and positive before termination. Exact algebra proves equality of (H.2) with the decomposed formulas (13.5); the decomposed form, rather than the fully expanded form, is used for inequalities.

H.2 Normalization of the radius

At a critical point set

$$T_k = \frac{y^k p_F^{(k)}(y)}{p_F(y)}.$$

Then $T_0 = 1$ and $T_1 = 0$. There is no $1/k!$ in this definition. Divide the recurrence at level m by $P_{2,m} R^{m+2}$, where $R = K_r \sqrt{Bd}$ and $K_r = 4096$. Under the exact bounds (14.1), the contributions of P_0, P_1, and P_3 are bounded by fixed strict fractions of one. The remaining inequalities use only the certified caps on d/n, B/n, and D/n.

Set $\mathcal{M} = \max_{0 \le j \le d} |T_j|/R^j$. If $\mathcal{M} > 1$, choose a maximizing index k. Since $T_0 = 1$ and $T_1 = 0$, we have $k \ge 2$. Use the recurrence with $m = k-2$, isolate its $P_{2,k-2}T_k$ term, and divide by $P_{2,k-2}R^k\mathcal{M}$. Every other normalized neighbor is at most one by the definition of $\mathcal{M}$; if $k = d$, the higher neighbor is $T_{d+1} = 0$ by termination. The three coefficient budgets have sum strictly below one, contradicting maximality. Therefore

$$\max_{1 \le k \le d} |T_k|^{1/k} \le K_r \sqrt{Bd}.$$

This global maximum, rather than lower-index minimality, is what controls the higher neighbor T_{k+1}. Thus no boundary order is silently omitted.

I Six-node interpolation and stability

I.1 The complex line-segment recursion

For holomorphic f on an open convex set and a, b in that set, complex FTC along the segment gives

$$f(b) - f(a) = (b-a) \int_0^1 f'(a + t(b-a))\, \mathrm{d}t. \tag{I.1}$$

Apply (I.1) recursively to Newton divided differences. After six steps, every evaluation point is a convex combination of $z_0, \ldots, z_5, z$. A stick-breaking parametrization of the standard six-simplex produces the weight

$$(1-u_0)^5(1-u_1)^4(1-u_2)^3(1-u_3)^2(1-u_4).$$

Its integral over $[0,1]^6$ is

$$\frac{1}{6}\frac{1}{5}\frac{1}{4}\frac{1}{3}\frac{1}{2} = \frac{1}{720}.$$

If the six node values vanish, the Newton formula therefore reduces to

$$f(z) = \prod_{j=0}^{5}(z - z_j) \int_{\Delta_6} f^{(6)}(\zeta(t))\, \mathrm{d}t,$$

which proves Lemma 15.1. Convexity is used exactly once: it keeps $\zeta(t)$ inside the derivative-bound domain.

I.2 Why the sixth match gives the exponent

For E_F, the nodes are $0, 1, \ldots, 5$. On the thickened critical-point domain, projection to $[0, d]$ and $\rho \geq d$ give $|z - j| \leq 3\rho$. Consequently

$$\begin{aligned} |E_F(z)| &\leq \frac{C_{B6}(3\rho)^6}{720n^5 \log(n+2)} \\ &\leq C_* \frac{d^3}{n^2 \log(n+2)}, \end{aligned}$$

because $\rho = K_r\sqrt{Bd}$ and $B \leq 3n$. Five matched nodes would produce a fifth power of the localization radius and recover the older three-fifths scale. The gain to two-thirds is therefore exactly the extra matching condition together with one additional derivative estimate.

The logarithm branch is legitimate because the coefficient ratios are nonzero on the parameter and complex-domain boxes. Once $|E_F| \leq 1$, the segment identity

$$e^{E_F} - 1 = E_F \int_0^1 e^{tE_F} \, \mathrm{d}t$$

gives $|e^{E_F} - 1| \leq e|E_F|$; the final constant is enlarged to use the simpler bound $2|E_F|$ on the smaller certified disc. This converts the logarithmic interpolation estimate to the multiplier estimate required in Lemma 16.1.

At each critical point, Cauchy's formula on the circle of radius ρ turns the bound on $c_F - 1$ into a convergent derivative tail. The critical point radius of Proposition 14.1 supplies the matching bound on the model derivatives. Rouché on disjoint critical-point discs then preserves one simple zero in each gap. Positivity of coefficients and the constant-term convention fix the sign of those roots; the rescaling $y = -SX$ sends them to distinct negative roots of $\mathcal{J}^{d,n}$.

J Effectivity, dependencies, and logical boundary

J.1 The finite pre-cutoff range is empty

Let N_0 be an integer above both the analytic and explicit construction cutoffs and set

$$K_{\text{finite}} = N_0^2(N_0 + 2) + 1.$$

For natural $0 \leq n < N_0$, the elementary inequality $\log x \leq x - 1$ gives

$$n^2 \log(n+2) \leq n^2(n+1) < N_0^2(N_0+2) < K_{\text{finite}}.$$

If $d \geq 1$, then $K_{\text{finite}} \leq K_{\text{finite}} d^3$. Hence

$$K_{\text{finite}} d^3 > n^2 \log(n+2),$$

which is the negation of the wedge antecedent. Increasing the final constant above K_{finite} only strengthens this exclusion. Thus "finite absorption" here does not mean that finitely many unverified Jensen polynomials are being declared hyperbolic; it means that the stated implication has no antecedent below the analytic cutoff.

The Lean definition `finiteCutoffAbsorptionConstant` and the theorem whose source name is `not_twoThirdsWedge_finiteCutoffAbsorption` formalize this calculation over natural N_0, n, d. This closes the formerly paper-only pre-cutoff step.

J.2 Finite and analytic constants

Let $x_{\text{admissible}}$ be the smallest rational cap on d/n demanded by the branch box, the Jacobi intervals, the global-maximum budgets, and the complex-domain containment. Since $\log(n+2) \le n$ for $n \ge 2$, the wedge implies $d/n \le K^{-1/3}$. The exact finite ledger takes

$$K_{\text{geometry}} = \lceil x_{\text{admissible}}^{-3} \rceil + 1 = 522255954073519803473068032000000000000000001. \quad \text{(J.1)}$$

The size of (J.1) reflects deliberately coarse uniform inequalities; it is not a prediction of the true transition.

Only two analytic quantities remain symbolic:

1. N_{analytic}, the threshold after which the fixed-sector saddle, contour, and coefficient estimates all hold;

2. C_{B6}, the explicit constant in the uniform sixth-derivative bound (15.6).

Once those are read from the displayed paper estimates, the final proof may take the maximum of K_{geometry}, the threshold conversion, and the multiplier constant. Thus the theorem is effective even though this paper does not optimize or print a practically useful numerical K.

J.3 What is and is not kernel checked

The Lean development proves the concrete centered-xi theta/Mellin/omega identity, factor eight, quantitative moving-sector saddle, legal contour, Gaussian localization, infinite higher-mode suppression, Gamma/Stirling assembly, and uniform coefficient theorem through order six. It also proves the exact xi whole-box branch, transformed comparison, six coefficient matches, finite-free specialization, complete critical-point localization, and literal normalized radius estimate. The Phase-26 axiom audit covers 1,217 declarations and reports only standard logical principles: propositional extensionality, classical choice, and quotient soundness. The project contains no `sorry`, `admit`, project-specific axiom, or unsafe escape hatch on this proof surface.

The Phase-28 terminal audit additionally covers the effective and literal forms of Theorem 7.1 and its derivative consequence through order six. The Phase-29 Palomar package isolates these statements and has passed the repository's local fresh-kernel, source-fidelity, and mutation checks. Palomar's own Comparator and NanoDa services have not yet replayed the candidate, so this document makes no official Palomar-verification claim.

The Phase-30 endpoint constructs the final xi-specific sixth-multiplier interval certificate, instantiates the sign-transfer theorem, identifies the actual transformed xi Jensen polynomial, and proves the headline negative-root conclusion. Exact symbolic systems and Arb/ACB remain corroborating channels rather than proof premises. The mathematical statements imported from classical Jacobi theory, MMP, and MSS are typed inputs to the concrete headline theorem rather than re-proved in Lean.

The terminal repair adds two literal closures. First, `riemannXiJensenPolynomialObject_natDegree` proves that the degree is exactly d from the positive leading coefficient, and the corresponding root theorem rules out $d+1$ distinct zeros. Second, `riemannXiJensen_twoThirds_global_headline_exactly` performs the $n \geq N_0$ versus $n < N_0$ split in one declaration and returns exactly d distinct negative real zeros. Its only non-kernel mathematical parameters are the explicitly typed Jacobi, MMP, and MSS literature inputs above the cutoff.

The most heavily imported external results are the classical Jacobi matrix/root correspondence, the finite-free nonnegative-root and log-mesh theorems in the pinned MMP version, and the MSS maximum-root theorem. The source audit records exact versions and theorem numbers, translates every polynomial convention explicitly, and marks these results as imported rather than re-proved. This separation is essential: a verified adapter is not a formalization of the external theorem it adapts.

J.4 Review status

The available analytic and algebraic adversarial reports were produced by AI systems with separated prompts and independent-recalculation requirements. They have found material transcription errors during development, and those errors were corrected and protected by regression tests. They are useful evidence, but they are neither human nor peer review. The release materials state this limitation prominently so that a prospective reviewer can assign the appropriate evidentiary weight to every channel.

References

[1] Michael Griffin, Ken Ono, Larry Rolen, Jesse Thorner, Zachary Tripp, and Ian Wagner. Jensen polynomials for the Riemann xi-function. arXiv:1910.01227v3, 2020. Version 3, frozen by SHA-256 in the accompanying source audit.

[2] Michael Griffin, Ken Ono, Larry Rolen, and Don Zagier. Jensen polynomials for the Riemann zeta function and other sequences. *Proceedings of the National Academy of Sciences*, 116(23):11103–11110, 2019.

[3] Jonathan Holland. A new hyperbolicity wedge and a joint semicircle limit for jensen polynomials of Riemann's xi-function. arXiv:2608.08682v1, 2026. Version 1, frozen by SHA-256 in the accompanying source audit.

[4] Adam W. Marcus, Daniel A. Spielman, and Nikhil Srivastava. Finite free convolutions of polynomials. *Probability Theory and Related Fields*, 182:807–848, 2022. The maximum-root product inequality used here is Theorem 1.6.

[5] Andrei Martínez-Finkelshtein, Rafael Morales, and Daniel Perales. Real roots of hypergeometric polynomials via finite free convolution. *International Mathematics Research Notices*, 2024(16):11642–11687, 2024. Exact proposition text checked against arXiv:2309.10970v3.